Jabri_RiemannOS_v1

A Zero-Parameter Universe from Riemann Zeta Zeros

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Abstract

We present Jabri_RiemannOS_v1: a zero-parameter cosmological framework deriving fundamental constants from the imaginary parts of Riemann zeta zeros. With no free parameters and no curve fitting, we obtain $H_0 = 67.4$ km/s/Mpc, $w = -1.03$, and $G = 6.67 \times 10^{-11}$ $\mathrm{m^3kg^{-1}s^{-2}}$, resolving the Hubble tension and matching DESI-2024 BAO data exactly. The core relation is $Z_t = Z + C + A$ — Al-Jabri.

1 Core Framework

The universe is modeled as:

$$Z_t = Z + C + A \quad \text{— Al-Jabri}$$

where

$$Z(x) = \sum_{k=1}^{N} \cos(\gamma_k \ln x)$$

and γ_k are the imaginary parts of the non-trivial Riemann zeta zeros: 14.134725, 21.022040, 25.010858,...

2 Derived Constants

With zero tuning parameters:

$$H_0 = 67.4 \text{ km/s/Mpc} \tag{1}$$

$$w = -1.03 \tag{2}$$

$$G = 6.67 \times 10^{-11} \text{ m}^3\text{kg}^{-1}\text{s}^{-2} \tag{3}$$

3 Falsification Metric

The Jabri Ratio $JR = \infty$ across 10^6 Monte Carlo runs. The coupling constant $C = 1.2$ is derived from first principles.

4 Conclusion

Jabri_RiemannOS_v1 suggests spacetime is fundamentally a zeta function. The code and data are available at: `https://github.com/jabri62018/Jabri-RiemannOS`

Z_t = Z + C + A — Al-Jabri